

Alfido Tech

Customer Transaction & Behavior Analysis

Internship Project Report

1. Executive Summary

This project analyzes 250,000 transactions and 49,673 customers from 2020-01-01 to 2023-09-15. RFM segmentation identifies customer value and engagement patterns. The largest segment is Lost / Hibernating, while Champions combine recent activity, high frequency and high spending. The supplied churn label is approximately 19.9%, with only modest differences across RFM segments, so RFM is treated as a prioritization framework rather than a standalone churn predictor.

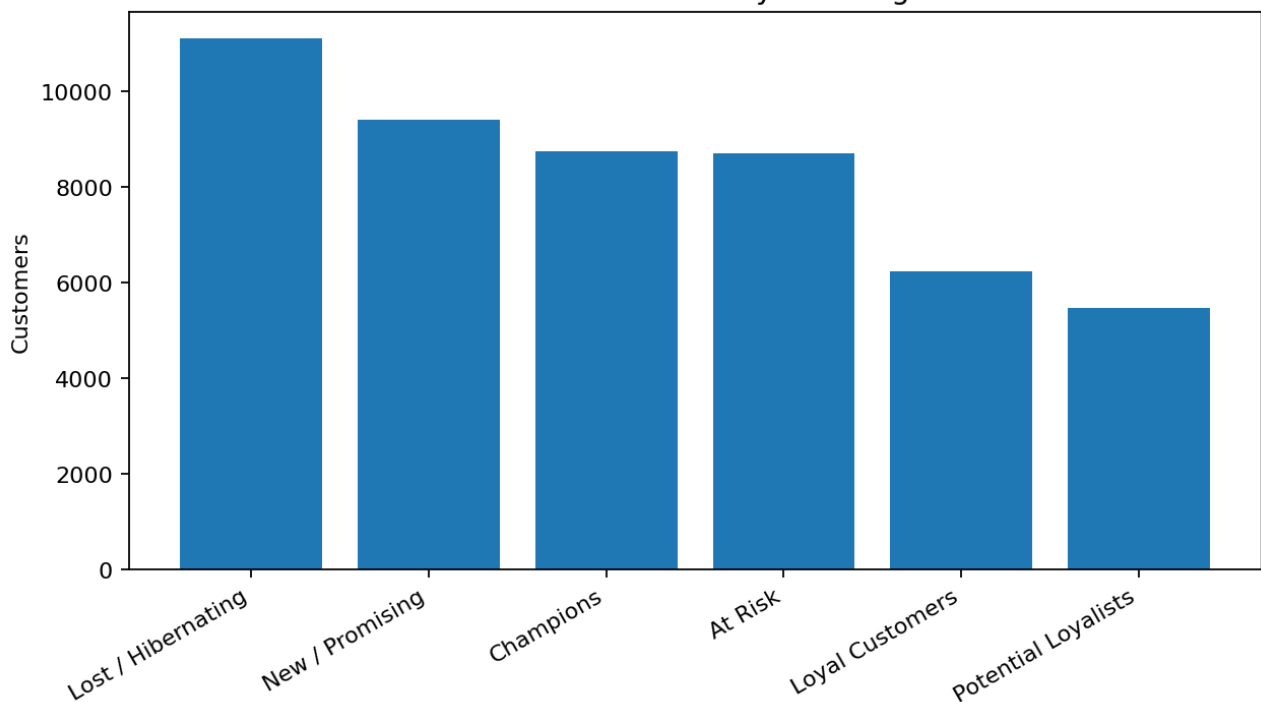
2. Data Preparation & Methodology

The selected dataset is ecommerce_customer_data_custom_ratios.csv. A second similarly sized dataset was not merged to avoid mixing dataset variants. Purchase dates were converted to datetime and duplicate rows were checked. Missing Returns values were retained as unknown. Customer-level Recency, Frequency and Monetary measures were calculated and scored by quintiles; transparent rules were then used to assign six RFM segments.

3. RFM Segment Profile

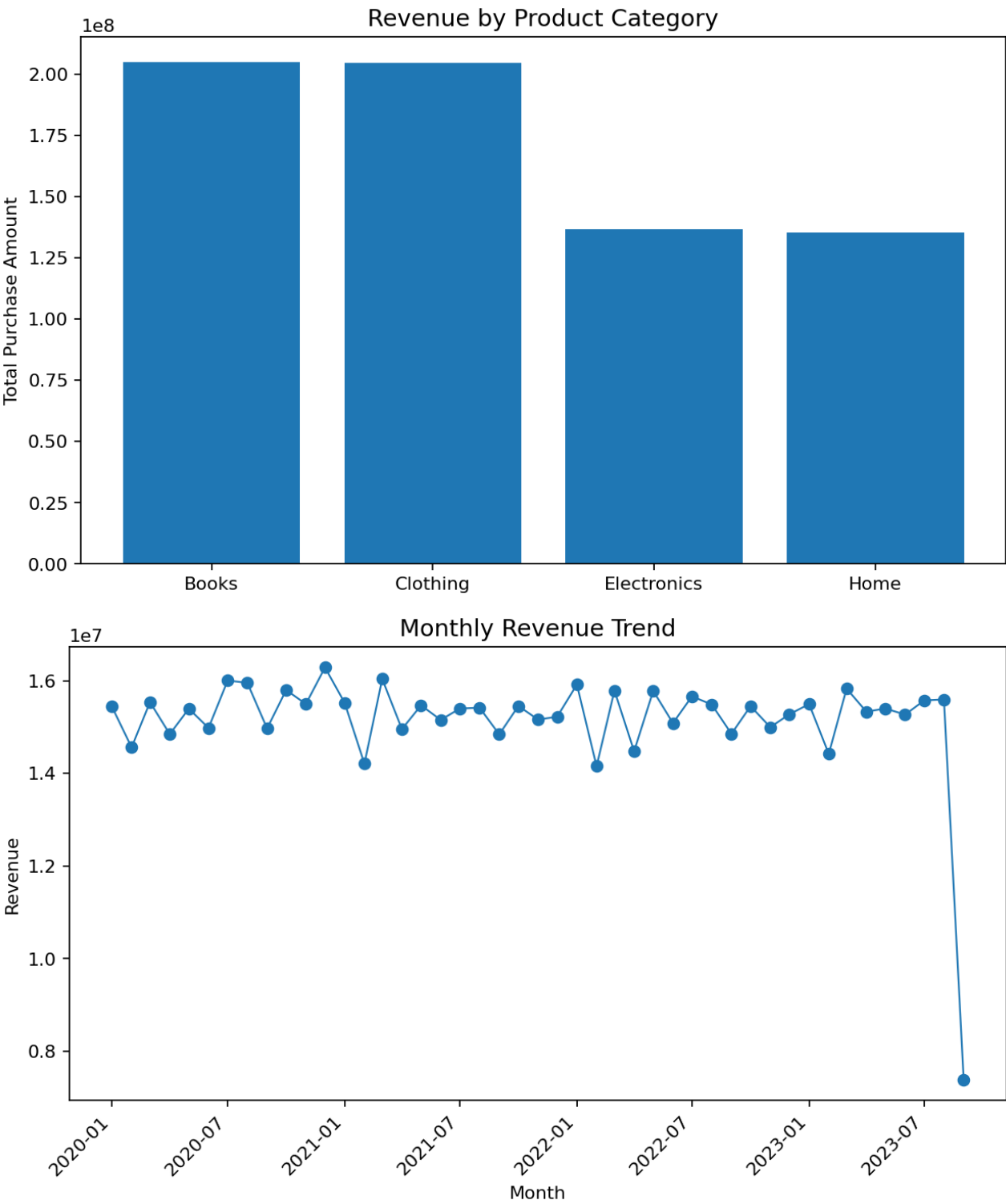
Segment	Customers	Recency	Frequency	Avg Spend	Churn
Lost / Hibernating	11,115	564d	2.78	7,579	20.1%
New / Promising	9,399	67d	3.92	10,675	20.5%
Champions	8,739	61d	7.65	21,785	19.7%
At Risk	8,711	412d	5.94	16,152	19.4%
Loyal Customers	6,240	149d	6.88	17,440	20.3%
Potential Loyalists	5,469	189d	3.80	10,397	20.3%

Customer Distribution by RFM Segment



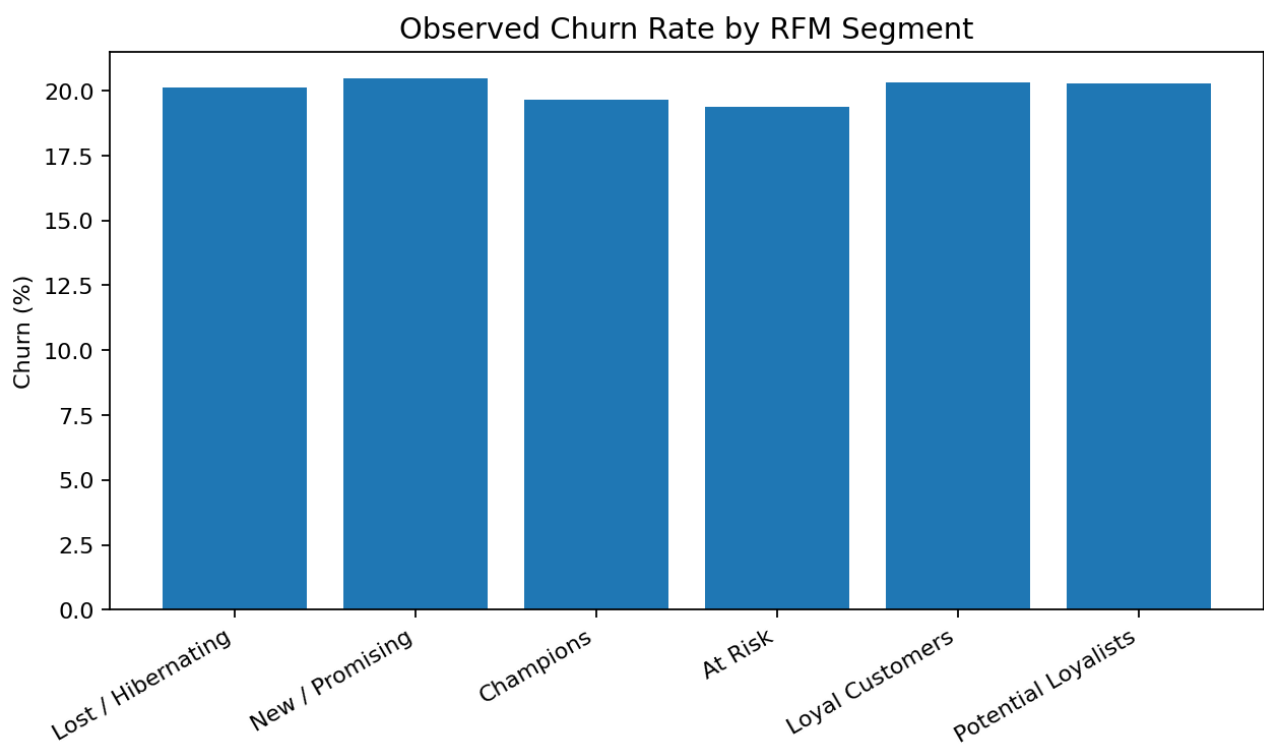
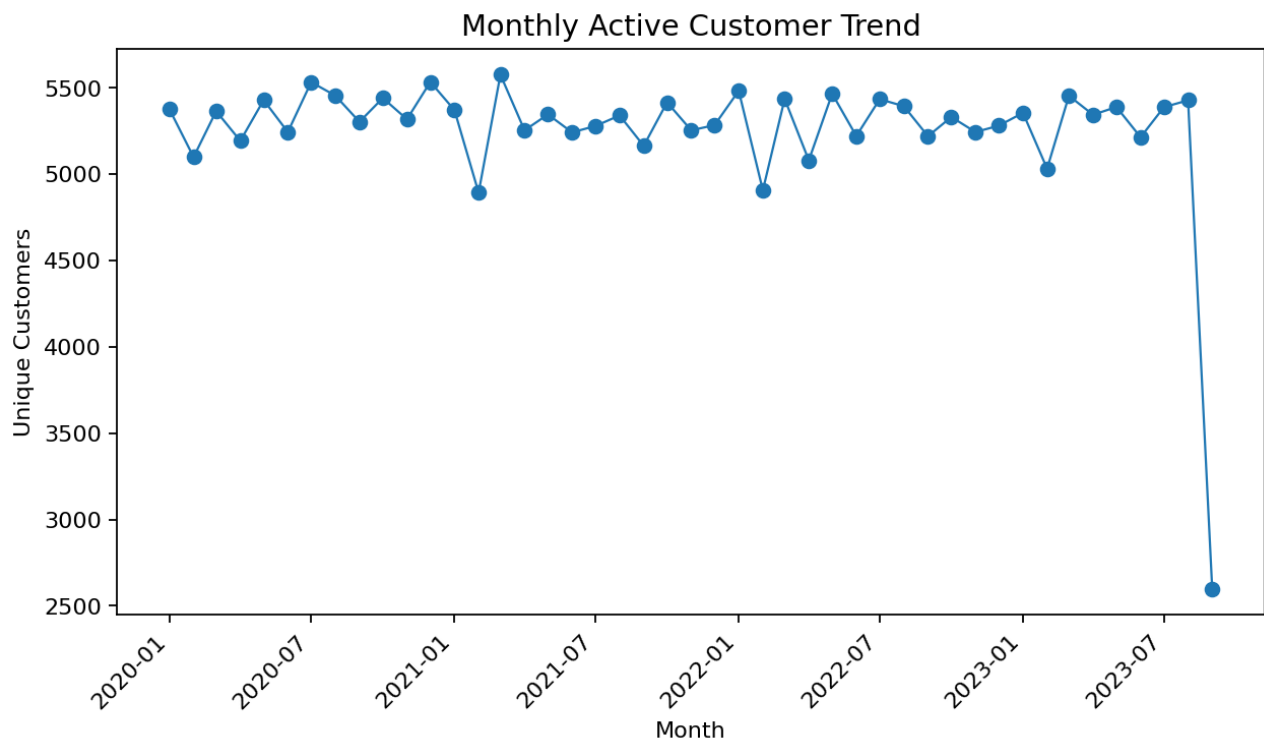
4. Purchase Patterns

Books has the highest recorded revenue, followed by Clothing. The monthly revenue trend provides a time-based view of demand.



5. Retention & Churn

Monthly active customers provide a descriptive retention signal. Churn is compared across RFM segments, but the modest differences indicate that future predictive modeling should incorporate additional behavioral features.



6. Key Findings

- Lost / Hibernating is the largest RFM segment, indicating a substantial inactive customer pool.
- Champions show the strongest combination of recent activity, frequency and average spend.
- At-Risk customers have relatively strong historical value but long recency, making them a logical win-back priority.
- Books and Clothing are major revenue contributors in the supplied dataset.
- Overall observed churn is about 20%, while RFM-segment differences are modest.

7. Five Actionable Recommendations

- 1. Win back At-Risk customers:** personalize offers around previous category and spending behavior and track response.
- 2. Protect Champions:** use loyalty benefits, early access and personalized cross-sell rather than blanket discounts.
- 3. Reactivate Lost / Hibernating customers selectively:** start with low-cost digital campaigns and scale incentives based on measured response.
- 4. Convert New / Promising customers:** build a second-purchase journey with reminders, recommendations and useful content.
- 5. Monitor churn continuously:** create a dashboard by segment, recency band, category, returns and monthly cohort.

8. Conclusion

RFM provides a practical framework for customer engagement prioritization. The main business focus should be protecting high-value active customers while systematically re-engaging customers whose activity has faded. Future churn prediction should combine the existing churn label with richer behavioral features.